

Combined computational and spectroscopic analyses of the interactions between ginger compounds and bovine type I collagen

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ABSTRACT

The stability of collagen is critical for meat quality; however, how ginger compounds act as collagen stabilizers is poorly understood. Molecular docking identified 24 compounds with binding affinities below -5.0 kcal/mol, involving Pro168, Tyr161, and Arg179 as key residues. Molecular dynamics simulations showed that collagen complexes with 3 major ginger compounds exhibited lower root mean square deviations than collagen, maintained by hydrogen bonds and hydrophobic interactions. Given that 6-shogaol is a dehydration product of 6-gingerol (6-GOL), 6-GOL and gingerenone A (GGA) were selected for experimental validation. Ultraviolet-visible absorption and fluorescence spectroscopy showed 6-GOL and GGA interacted with collagen. Fourier transform infrared spectroscopy revealed that both compounds altered the Amide bands, confirming strengthened hydrogen bonding. Furthermore, differential scanning calorimetry analysis demonstrated that both compounds enhanced the thermal stability of collagen. These findings demonstrate that 6-GOL and GGA stabilize bovine type I collagen, providing insights into their potential as collagen stabilizers.

1. Introduction

Mammals contain collagen as their primary structural protein which follows a repeating pattern of Gly-X-Y sequences (Mo et al., 2023; Moon, 2018). As the major protein in connective tissue, collagen significantly influences the texture of meat (Roy & Bruce, 2024). During thermal processing, collagen may undergo denaturation that significantly affects the tenderness of meat (Li, Ha, et al., 2022; Purslow, 2018). Within the collagen family, type I collagen is the most abundant, which plays a key role in the stability of meat (Mikolajczak et al., 2019). Therefore, enhancing the stability of type I collagen can improve the quality of meat. Notably, bovine type I collagen has become a focus in meat research due to its wide availability (Naomi et al., 2021).

In recent years, natural compounds have been used for regulating collagen stability. Studies have demonstrated that these compounds can enhance the structural stability of collagen by forming hydrogen bonds or covalent cross-links (Feng et al., 2019; Lucarini et al., 2020). Ginger (*Zingiber officinale*), as a typical edible and medicinal plant (Xu et al., 2022), serves as a widely used herbal medicine (Ballester et al., 2022). Research indicates that ginger is rich in phenolic compounds, including gingerols, shogaols, and gingerenones (Dalsasso et al., 2022; Rasheed,

2020; Xue et al., 2022), and these phenolic compounds possess various bioactive properties, including antioxidant, anticancer, anti-inflammatory, anti-angiogenic, and anti-atherosclerotic effects (Habib et al., 2008). In addition, both ginger powder and ginger extract have been shown to act as natural stabilizers (Mehrotra et al., 2024; Neba et al., 2020; Tiencheu et al., 2021). However, the molecular mechanisms by which these ginger compounds potentially enhance the stability of collagen and thereby improve meat quality have not been elucidated.

To bridge this gap and provide a rational screening pipeline, in silico biology is a rapidly developing field, integrating modeling to predict and interpret biological functions (Vidal-Limon et al., 2022). Molecular docking and molecular dynamics (MD) simulations are widely employed for biomolecular interactions. Molecular docking is used to predict the binding patterns between ligands and receptors (Santos et al., 2019). Based on the docking results, MD simulations can be used to investigate the dynamic stability of the protein-ligand complexes (Liu et al., 2022). Nevertheless, it is still necessary to carefully consider some inherent limitations in the prediction receptor model, such as the static characteristics of the predicted side chain configuration and the lack of natural post-translational modifications, which are of great biological

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significance for stabilizing the structure of collagen (Li, Zhang, et al., 2022; Wu et al., 2026). It is because of these specific limitations that the subsequent MD simulation is adopted, which aims to enable the structure of the receptor-ligand complex to be dynamically relaxed and adaptable, and to verify its binding stability.

Overall, this study investigated the interaction mechanism between ginger compounds and bovine type I collagen using computational simulation methods and multispectral analysis to provide insights into their potential as natural collagen stabilizers. We identified 30 compounds in ginger from multiple literature sources and further screened them using molecular docking (Attallah et al., 2025; Garza-Cadena et al., 2023; Matin et al., 2024; Yang et al., 2024). Subsequently, 3 major ginger compounds, 6-shogaol (6-SHO), gingerenone A (GGA), and 6-gingerol (6-GOL), were singled out for 100 ns MD simulations. Finally, because 6-shogaol is a dehydration derivative of 6-gingerol (Jin et al., 2024; Torquato et al., 2017), the study employed ultraviolet–visible (UV–vis) absorption spectroscopy, fluorescence spectroscopy (FS), Fourier transform infrared spectroscopy (FTIR), and differential scanning calorimetry (DSC) to analyze the collagen binding properties of GGA and 6-GOL, and their effects on collagen stability.

2. Materials and methods

2.1. Materials

6-GOL and GGA (both $\geq 98\%$ purity) were purchased from Solarbio Science and Technology Co., Ltd. (Beijing, China). Bovine type I collagen was obtained from Yuan Ye Biotechnology Co., Ltd. (Shanghai, China). KBr was purchased from Macklin Biochemical Technology Co. Ltd. (Shanghai, China). All chemical reagents used in this experiment were of analytical grade and were stored and used strictly according to the manufacturers' instructions.

2.2. Computational methods

2.2.1. Molecular docking

To investigate the binding affinity and potential binding patterns between bioactive components derived from ginger and bovine type I collagen, molecular docking analysis was performed using AutoDock Vina (1.1.2). First, 30 ginger compounds were screened through literature review, and their 3D structures were obtained from PubChem as ligands. The 3D structures of the receptor, bovine type I collagen, were acquired via AlphaFold Protein Structure Database (AF-Q95ND8-F1 model), which exhibited high quality with an average pLDDT of 88. Energy minimization was performed on the 30 ligands using the MMFF94 force field within Open Babel 3.0.0. All ligand and receptor structures were preprocessed using MGLTools (v1.5.7) and converted to PDBQT format for docking. The blind docking strategy was used with a $33.64 \times 26.91 \times 60.55$ Å grid box which ensured sufficient spaced to cover the entire surface of the collagen (Cai et al., 2021). Additional parameters were set as follows: exhaustiveness = 8, num_modes = 5. Subsequently, the molecular docking results were visualized using PyMOL (v2.1) and LigPlot⁺ (v2.2) to identify interactions between ligands and proteins (Laskowski & Swindells, 2011; Wallace et al., 1995).

2.2.2. MD simulations

Based on molecular docking results, 6-GOL-collagen, 6-SHO-collagen, and GGA-collagen were selected for MD simulations. All simulations were performed using gromacs 2025.1. The collagen was processed using the AMBER99SB force field (M. Abraham et al., 2025; Hornak et al., 2006). Ligand molecules were processed using the Sobotop tool based on the AMBER General All-atom Force Field. The complex system was placed in a triclinic box, solvated with the SPC water model, and supplemented with an appropriate amount of Na⁺/Cl⁻ ions to maintain electrical neutrality. The energy minimization process consisted of two stages which started with steepest descent optimization

before switching to conjugate gradient method for achieving stability. The system required 200 ps of NVT equilibration at 300 K before it underwent 200 ps of NPT equilibration to reach complete temperature and pressure equilibrium using a Parrinello-Rahman barostat at 1.0 bar. Long-range electrostatic interactions were treated using the Particle Mesh Ewald (PME) method, with short-range van der Waals and electrostatic cutoffs both set to 1.0 nm. All bonds involving hydrogen atoms were constrained using the LINCS algorithm. A 100 ns production simulation was then performed for each system with an integration time step of 2 fs. Each complex system was simulated independently as a single continuous run (Abraham et al., 2015; Hess et al., 2008; Pronk et al., 2013). Using DulvyTools package (v0.6.0) (CharlesHahn, 2024), we analyzed the root mean square deviation (RMSD), root mean square fluctuation (RMSF), radius of gyration (Rg), hydrogen bonds number and solvent accessible surface area (SASA) to evaluate the conformational dynamics and binding stability of the complex.

2.3. Multispectral analyses

2.3.1. Preparation of 6-GOL/GGA-bovine type I collagen complexes

Dissolve 6-GOL and GGA separately in 1 mL of ethanol, then dilute with ultrapure water to achieve initial concentrations of 20, 40, 60, 80, and 100 µg/mL, respectively, and the pH was maintained at neutral. Subsequently, each concentration of 6-GOL or GGA solution was mixed with an equal volume of bovine type I collagen solution (initial concentration 100 µg/mL). For the control group, collagen was mixed with an equal volume of the corresponding solvent vehicle. After thorough mixing of each system, the mixtures were incubated at 25 °C for 30 min to ensure complete complex formation. The final protein concentrations were 50 µg/mL.

2.3.2. UV–vis absorption spectroscopy

UV–vis absorption spectroscopy were acquired for the 6-GOL/GGA-bovine type I collagen complex solutions prepared in Section 2.3.1 using a UV–visible spectrophotometer. The scanning wavelength range was from 200 to 300 nm, with collagen solutions at corresponding concentrations serving as blank controls.

2.3.3. FS analysis

The FS of the 6-GOL/GGA-bovine type I collagen complex solution prepared in Section 2.3.1 were measured using a fluorescence spectrometer. The excitation wavelength was set at 280 nm, and the emission spectra of 6-GOL/GGA-bovine type I collagen complex solutions at different concentrations were recorded within the 300–500 nm range. Both excitation and emission slit widths were maintained at 5 nm. Collagen solutions at corresponding concentrations served as blank controls.

2.3.4. FTIR analysis

The samples were first frozen at −20 °C for 9 h, then transferred to −80 °C for an additional 24 h. After 72 h of freeze-drying, the corresponding lyophilized powders were obtained. An appropriate amount of the lyophilized powder was mixed with KBr and thoroughly ground before being pressed into transparent pellets. Spectra were then acquired at room temperature using a Fourier Transform Infrared Spectrometer within the range of 400–4000 cm^{−1}. Each spectrum was collected with 128 scans at a resolution of 4 cm^{−1}.

2.3.5. DSC analysis

Solutions of collagen-6-GOL complexes and collagen-GGA complexes were prepared separately using the method described in Section 2.3.1. The thermal stability of each sample was then evaluated using differential scanning calorimetry. The tests were conducted in a nitrogen atmosphere at a heating rate of 5 °C/min, with a scanning range of 30 °C to 120 °C.

3. Results

3.1. Molecular docking

Molecular docking was employed to screen the binding potential of ginger compounds. The binding affinities of the 30 screened compounds with bovine type I collagen ranged from -4.6 to -6.7 kcal/mol (Table 1). Notably, 24 compounds exhibited binding affinities stronger than -5.0 kcal/mol. The binding energy values below the threshold of -5.0 kcal/mol indicate good binding activity (Lin et al., 2025; Liu et al., 2024; Zhang et al., 2024). Zerumbone (ZER) showed the strongest affinity (-6.7 kcal/mol), followed closely by Gingerenone A (GGA, -5.7 kcal/mol) and 6-Shogaol (6-SHO, -5.6 kcal/mol). 6-Gingerol (6-GOL), the major pungent component, showed a binding affinity of -5.0 kcal/mol. Visual analysis of the docking poses revealed that these compounds bound to specific pockets on the collagen surface (Table 2, Fig. 1 and Fig. 2). The interaction is driven mainly by hydrogen bonds and hydrophobic interactions. Residues Arg179, Tyr161, and Pro168 were identified as conserved binding sites, participating in interactions with almost all top-ranking compounds. For instance, Arg179 and Tyr161 frequently acted as hydrogen bond donors or acceptors, anchoring the ligands to the collagen.

3.2. MD simulation

In order to further study the structural stability of the complex, a 100 ns molecular dynamics simulation was carried out. The results showed that all the complexes reached equilibrium during the simulation process (Fig. 3). Compared with apo collagen, the RMSD of all complexes were lower than that of apo collagen. Among them, the RMSD value of 6-SHO-collagen complex is the lowest. RMSF analysis further revealed that while the 6-GOL-collagen complex exhibited relatively higher residue

Table 1
The binding affinities of 30 active compounds in ginger for bovine type I collagen.

No.	Compounds	Abbreviation	Affinity (kcal/mol)
1	Zerumbone	ZER	-6.7
2	Curcumin	CUR	-6.6
3	1-(4-hydroxy-3-methoxyphenyl)-7-(3,4-dihydroxyphenyl)-4e-en-3-heptanone	DHP-HMH	-6.1
4	1,7-bis(3,4-dihydroxyphenyl)heptan-3-yl acetate	DHPHA	-6.0
5	Gingerenone A	GGA	-5.7
6	Gingerdione	GGD	-5.7
7	8-Shogaol	8-SHO	-5.7
8	6-Shogaol	6-SHO	-5.6
9	Gingerdiol	GDL	-5.5
10	Zingerone	ZGO	-5.5
11	Hexahydrocurcumin	HHC	-5.4
12	Borneol	BOR	-5.4
13	8-Gingerdiol	8-GDL	-5.4
14	10-Gingerol	10-GOL	-5.3
15	Cineole	GIN	-5.3
16	10-Dehydrogingerdione	10-DGDO	-5.2
17	8-Paradol	8-PAD	-5.2
18	Dehydrogingerdione	DHGDO	-5.2
19	10-Gingerdione	10-GDO	-5.2
20	(+)-Alpha-Pinene	α -PIN	-5.2
21	Beta-Phellandrene	β -PHE	-5.1
22	Paradol	PAD	-5.1
23	8-Gingerol	8-GOL	-5.1
24	1-Dehydro-10-Gingerdione	1-DGDO	-5.0
25	Neral	NER	-5.0
26	6-Gingerol	6-GOL	-5.0
27	10-Shogaol	10-SHO	-4.9
28	Citral	CIT	-4.7
29	Linalool	LIN	-4.6
30	12-Gingerol	12-GOL	-4.6

Table 2
Molecular docking interactions of nine ginger compounds with bovine type I collagen: total interacting residues and hydrogen-bond details.

Compounds	Total interacting residues	Residues Num.	Hydrogen-bond residues	H-bonds Num.
GGA	Ile263, Gln158, Ala172, Ser216, Glu160, Pro168, Leu175, Pro261, Arg179, Tyr161, Phe159	10	Arg179, Tyr161, Phe159	2
6-SHO	Ile263, Ala172, Gln158, Pro261, Pro168, Phe159, Glu160, Leu175, Arg179, Tyr161, Ser216	11	Arg179, Tyr161, Ser216	2
6-GOL	Ile263, Gln158, Glu160, Pro168, Leu175, Pro261, Tyr161, Phe159, Arg179, Ser216	10	Arg179	1
ZER	Glu160, Pro168, Leu175, Pro261, Tyr161, Phe159, Arg179, Ser216, Ala172	9	Arg179	1
CUR	Glu160, Pro168, Leu175, Pro261, Tyr161, Ser216, Ala172, Gln158, Ile263, Phe159	11	Gln158, Tyr161	2
DHP-HMH	PRO261, TYR161, VAL171, LEU175, ALA172, PRO168, SER216, IIE263, Phe159, Arg179, Glu160, GLN158	12	Phe159, Arg179, Glu160	3
DHPHA	Pro261, Val171, Leu175, Ala172, Ser216, Arg179, Glu160, Ile263, Phe159, Pro168, Tyr161	11	Pro168, Tyr161	2
GGD	Gln158, Ser216, Pro261, Leu175, Ala172, Val171, Ile263, Pro168, Phe159, Glu160, Arg179, Tyr161	12	Arg179, Tyr161	2
8-SHO	Pro168, Ile263, Glu160, Gln158, Phe159, Ala172, Leu175, Pro261, Ser216, Arg179, Tyr161	11	Arg179, Tyr161	2

Note: ZER, Zerumbone; CUR, Curcumin; DHP-HMH, 1-(4-hydroxy-3-methoxyphenyl)-7-(3,4-dihydroxyphenyl)-4e-en-3-heptanone; DHPHA, 1,7-bis(3,4-dihydroxyphenyl)heptan-3-yl acetate; GGD, Gingerdione; 8-SHO, 8-shogaol.

flexibility, the RMSF values of the other 2 complexes were lower than those of apo collagen. In terms of the Rg value, the Rg value of apo collagen was slightly lower than that of the complex. The addition of the ligand leads to a slight increase in the Rg value, which indicates that the structure was slightly expanded to accommodate the ligand upon ligand binding, although the overall compactness remains basically unchanged. Throughout the MD simulations, all three complexes maintained stable hydrogen bond networks. Compared to apo collagen, the SASA values of these complexes showed slight increases. Additionally, Fig. 4 illustrates the changes in secondary structure for collagen and its complexes during simulation. After 75 ns, distinct trends emerged in how different ligands influenced collagen's secondary structure. Specifically, GGA binding increased α -helix content while slightly decreasing 3-helix and loop content. The addition of 6-GOL and 6-SHO similarly reduced 3-helix content; however, unlike GGA, these compounds increased loop region content.

3.3. UV-vis absorption spectroscopy and FS analysis

UV-vis absorption spectroscopy analysis revealed characteristic absorption peaks near 205 nm for bovine type I collagen (Fig. 5A). Following the addition of 6-GOL and GGA, collagen's UV-vis absorption spectroscopy exhibited a prominent absorption band near 205 nm, with absorbance increasing as 6-GOL or GGA concentration rose, exhibiting a concentration-dependent manner accompanied by a slight red shift. The UV-vis absorption spectroscopy of pure 6-GOL and GGA were included

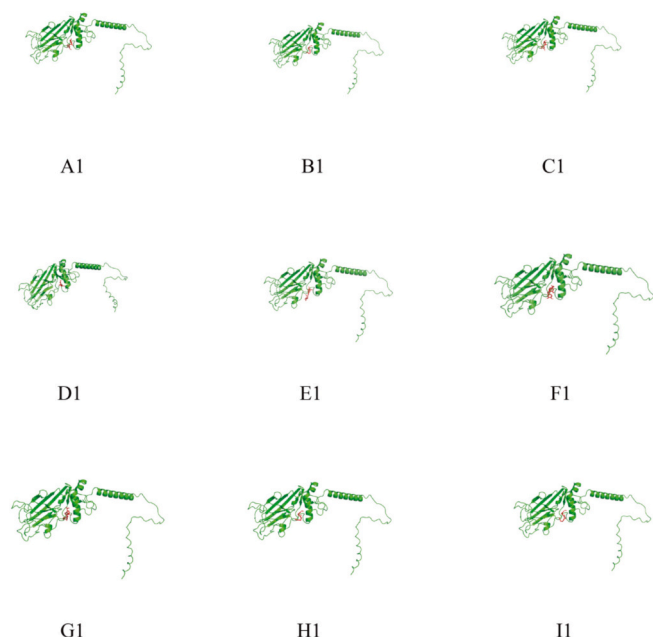


Fig. 1. 3D molecular docking conformations of nine ginger compounds with bovine type I collagen. (A1) GGA, (B1) 6-SHO, (C1) 6-GOL, (D1) ZER, (E1) CUR, (F1) DHP-HMH, (G1) DHPHA, (H1) GGD, (I1) 8-SHO.

as controls. Since our target protein (collagen) lacks significant intrinsic absorption in the 270–300 nm range, we selected the characteristic absorption peak of the ligand at ~ 279 nm for comparison. By comparing the spectra of the complex and pure small molecule solutions at the same concentration, we observed distinct changes in absorbance following binding. Taking a concentration of 20 $\mu\text{g/mL}$ as an example: At 279 nm, the absorbance of the 6-GOL complex was 0.101, which was higher than the intrinsic absorbance of pure 6-GOL at the same concentration (0.078), exhibiting a hyperchromic effect. Conversely, the absorbance of the GGA complex was 0.154, which was lower than the intrinsic absorbance of pure GGA at the same concentration (0.172), exhibiting a hypochromic effect.

FS was further employed to characterize the formation of collagen-ligand complexes (Fig. 5B). Collagen exhibited a characteristic fluorescence emission peak at 309 nm upon excitation at 280 nm. The addition of 6-GOL or GGA (20–100 $\mu\text{g/mL}$) caused a regular increase in fluorescence intensity, indicating an interaction between collagen and the compounds.

3.4. FTIR

As illustrated in Fig. 6, the FTIR spectra of native bovine type I collagen and its complexes with 6-GOL/GGA exhibited characteristic absorption bands associated with the polypeptide backbone. The prominent peaks were identified as Amide A ($\sim 3324\text{ cm}^{-1}$, N–H stretching), Amide B ($\sim 2926\text{ cm}^{-1}$, asymmetric CH_2 stretching), Amide I ($\sim 1650\text{ cm}^{-1}$, C=O stretching), Amide II ($\sim 1540\text{ cm}^{-1}$, N–H bending coupled with C–N stretching), and Amide III ($\sim 1240\text{ cm}^{-1}$, C–N stretching and N–H bending). Comparative analysis showed that the addition of 6-GOL and GGA altered the intensity and position of the Amide I and Amide II bands. Specifically, the Amide I band in the complexes shifted to a lower wavenumber (red shift) relative to native collagen. In addition, the collagen–6-GOL spectrum (red line) exhibited a noticeable increase in absorbance intensity compared with the control.

3.5. DSC analysis

As shown in Fig. 7, the thermal denaturation temperature of pure

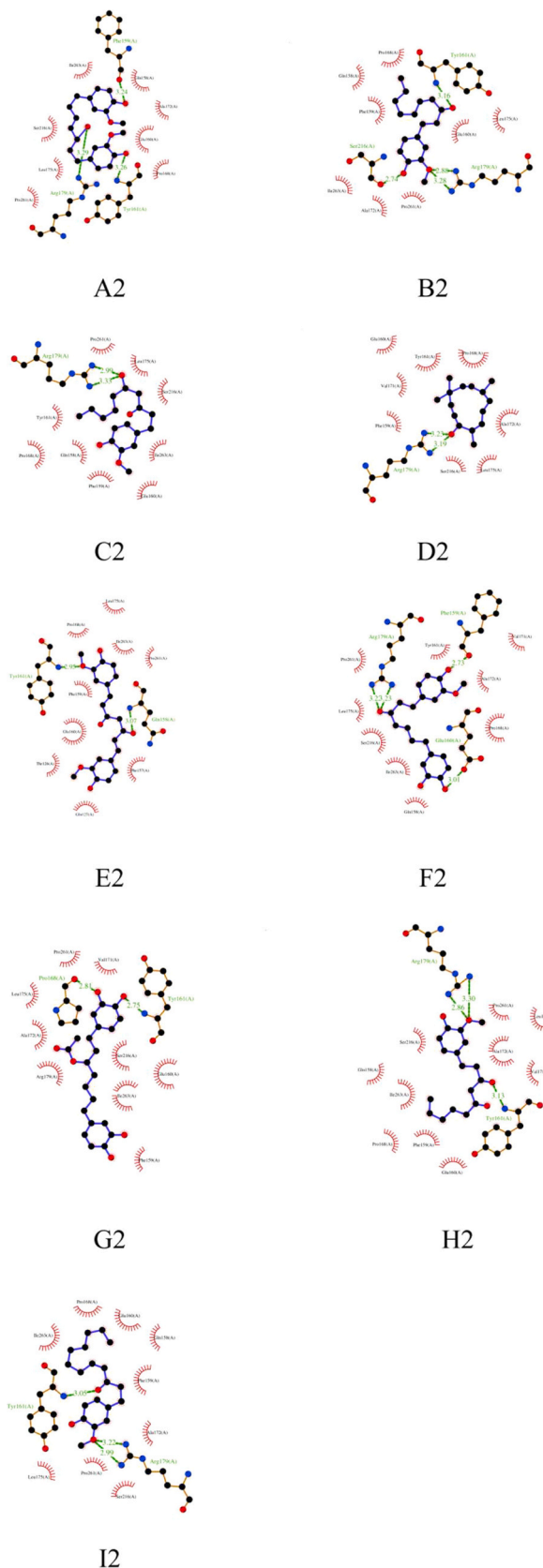


Fig. 2. 2D molecular docking conformations of nine ginger compounds with bovine type I collagen. (A2) GGA, (B2) 6-SHO, (C2) 6-GOL, (D2) ZER, (E2) CUR, (F2) DHP-HMH, (G2) DHPHA, (H2) GGD, (I2) 8-SHO.

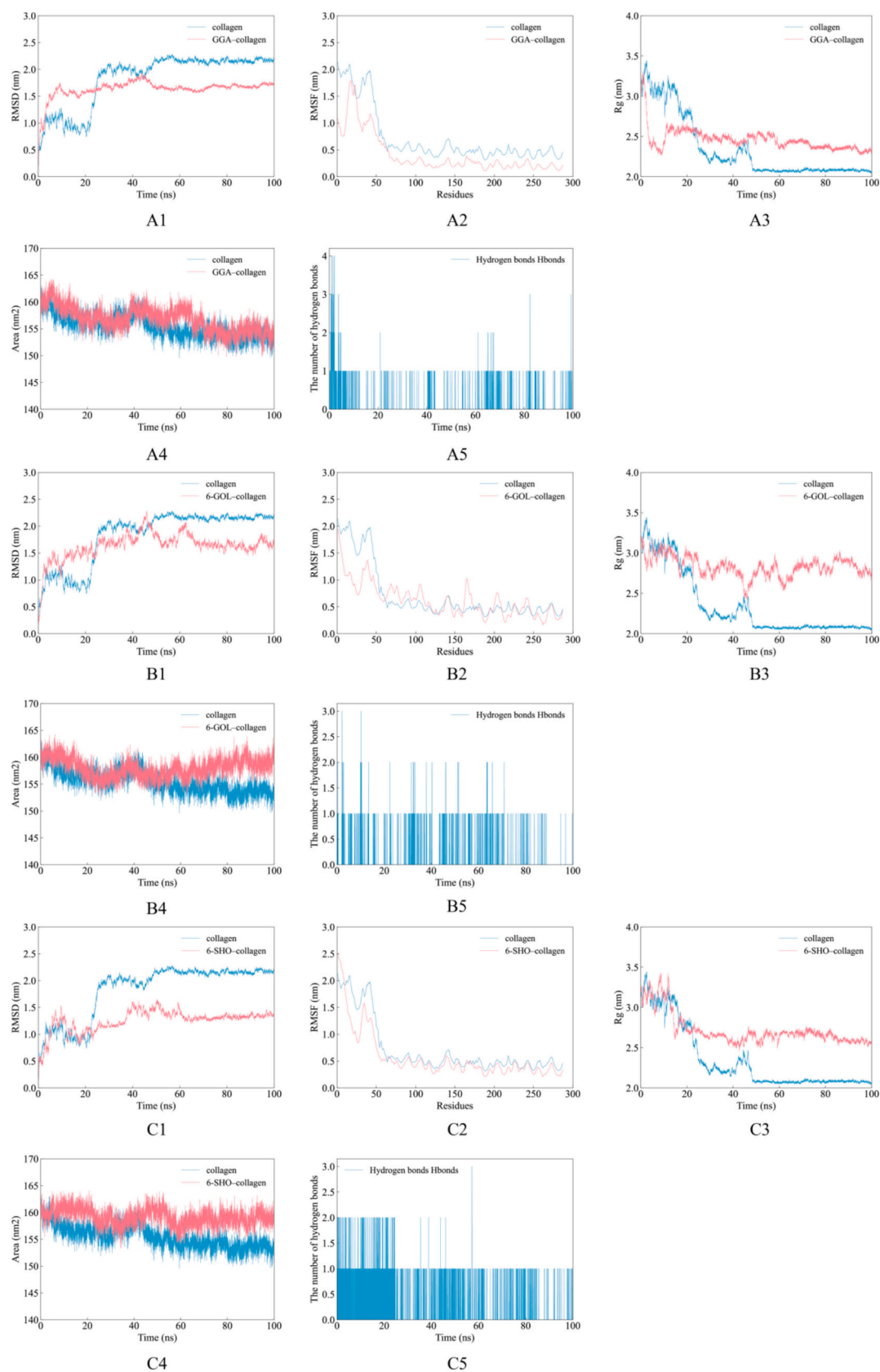


Fig. 3. Structural dynamics of collagen-ligand complexes. (A1–C1) RMSD, (A2–C2) RMSF, (A3–C3) Rg, (A4–C4) SASA, and (A5–C5) the number of hydrogen bonds. Note: A, B and C correspond to GGA, 6-GOL, and 6-SHO, respectively.

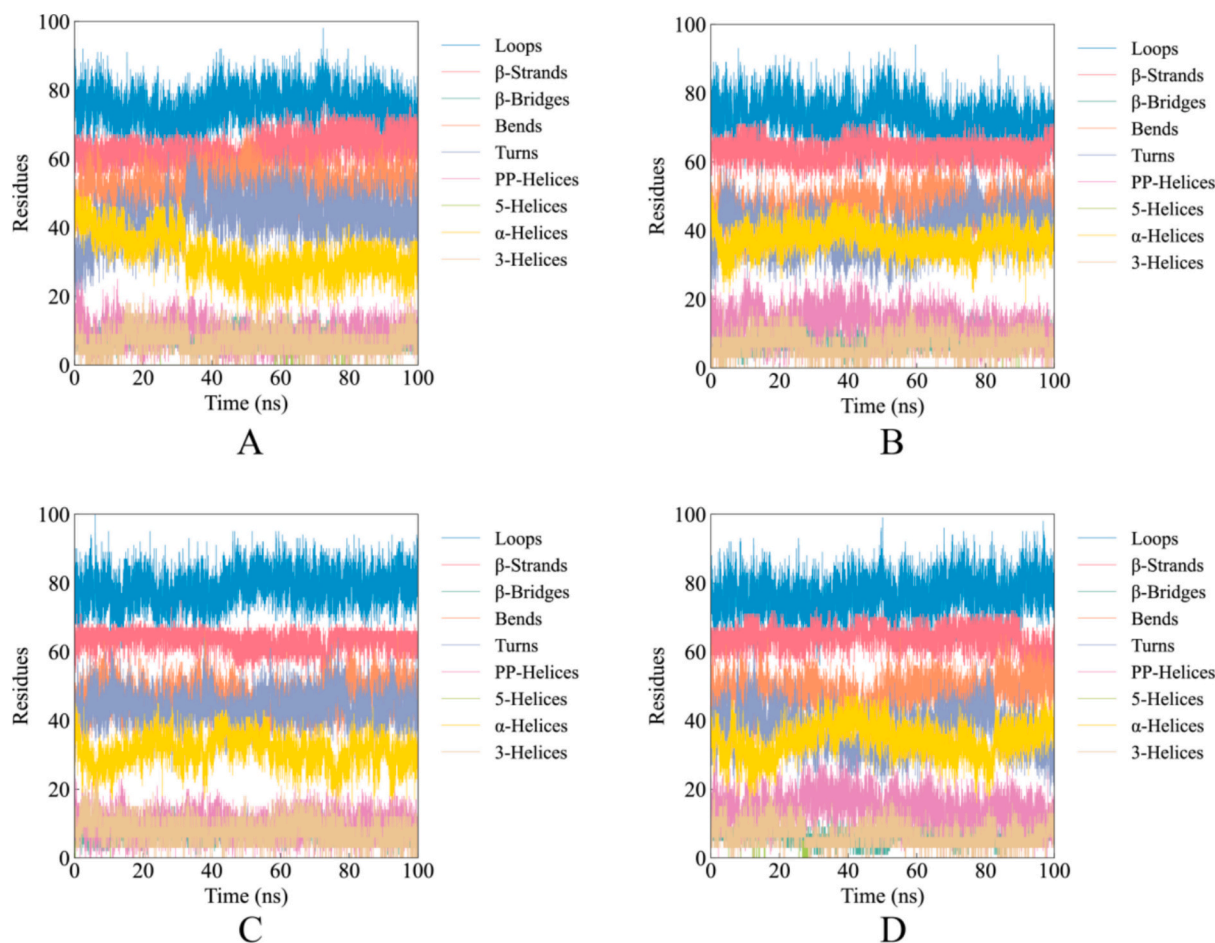


Fig. 4. Secondary structure trends of collagen and collagen-ligand complexes during MD simulation. (A) Collagen, (B) GGA-collagen, (C) 6-GOL-collagen, (D) 6-SHO-collagen.

collagen was 101.29 °C. After binding with the two ginger-derived compounds, the thermal denaturation temperatures of the complexes increased: the thermal denaturation temperature rose to 103.05 °C upon addition of GGA, and to 104.19 °C upon addition of 6-GOL.

4. Discussion

Collagen is the main structure of connective tissue, and its thermal stability is critical determinants of meat texture (Liu & Xing, 2023). During cooking, the thermal denaturation of collagen leads to shrinkage and fluid loss (Choe & Kim, 2019). The process of thermal denaturation in collagen requires stabilization because it directly affects both meat tenderness and production yield. The present study shows that 6-GOL and GGA from ginger function as natural stabilizers for collagen.

The molecular docking process of 30 ginger compounds showed that 24 compounds demonstrated favorable binding thermodynamics, with affinity values which were below −5.0 kcal/mol (Priyadarshini et al., 2025). Subsequently, a visual analysis of the docking results was performed for the 9 compounds. The results revealed that these compounds formed hydrogen bonds and hydrophobic interactions with collagen. The study found that Pro168, Tyr161 and Arg179 acted as recurring essential binding sites across multiple ligand interactions. The hydrogen bond donor role in most complexes belonged to Arg179 and Tyr161 among all the identified residues. The strong binding of ginger compounds to collagen can be attributed to the formation of hydrogen bonds with collagen which maintains collagen structure and improves meat tenderness (Bai et al., 2025). The research results match previous studies which investigated how polyphenols interact with collagen molecules

(Madhan et al., 2005).

The dynamic stability of these interactions required 100 ns MD simulations to be performed on the three primary collagen-ligand complexes which included Collagen-GGA, Collagen-6-SHO and Collagen-6-GOL. The RMSD values from all three complexes displayed lower values than apo collagen values which showed that ligand binding increased collagen structure stability (Haryini & Doss, 2025). The RMSF analysis showed that all complexes except the 6-GOL-collagen complex displayed RMSF values lower than apo collagen values thus indicating better local structural rigidity (Ma et al., 2023). The structural variations among the ligands significantly modulated these dynamics; specifically, the extended aliphatic chain and multiple hydroxyl groups in GGA served as multiple anchoring points (Sun et al., 2023), effectively stabilizing the complex and resulting in the lowest RMSF values (Yoosefian et al., 2025). The structural compactness analysis showed that all three complexes experienced only a small increase in their Rg values, indicating a minor spatial accommodation of the ligands without unraveling the core triple-helix structure. When ligands bound which indicated their structures became slightly more compact (Jibon et al., 2025). The secondary structure analysis showed that the GGA-collagen complex developed more α-helix structures while its 3-helix and loop regions became slightly smaller during the 75 ns simulation. The implementation of 6-GOL and 6-SHO led to a decrease in 3-helix structure but the ring areas gained new content which showed that particular ligands triggered distinct structural changes which impacted collagen stability (Derewenda & Godzik, 2017). The research into system interactions revealed that all 3 complexes formed enduring hydrogen bond networks yet the 6-SHO complex maintained the most stable hydrogen bond

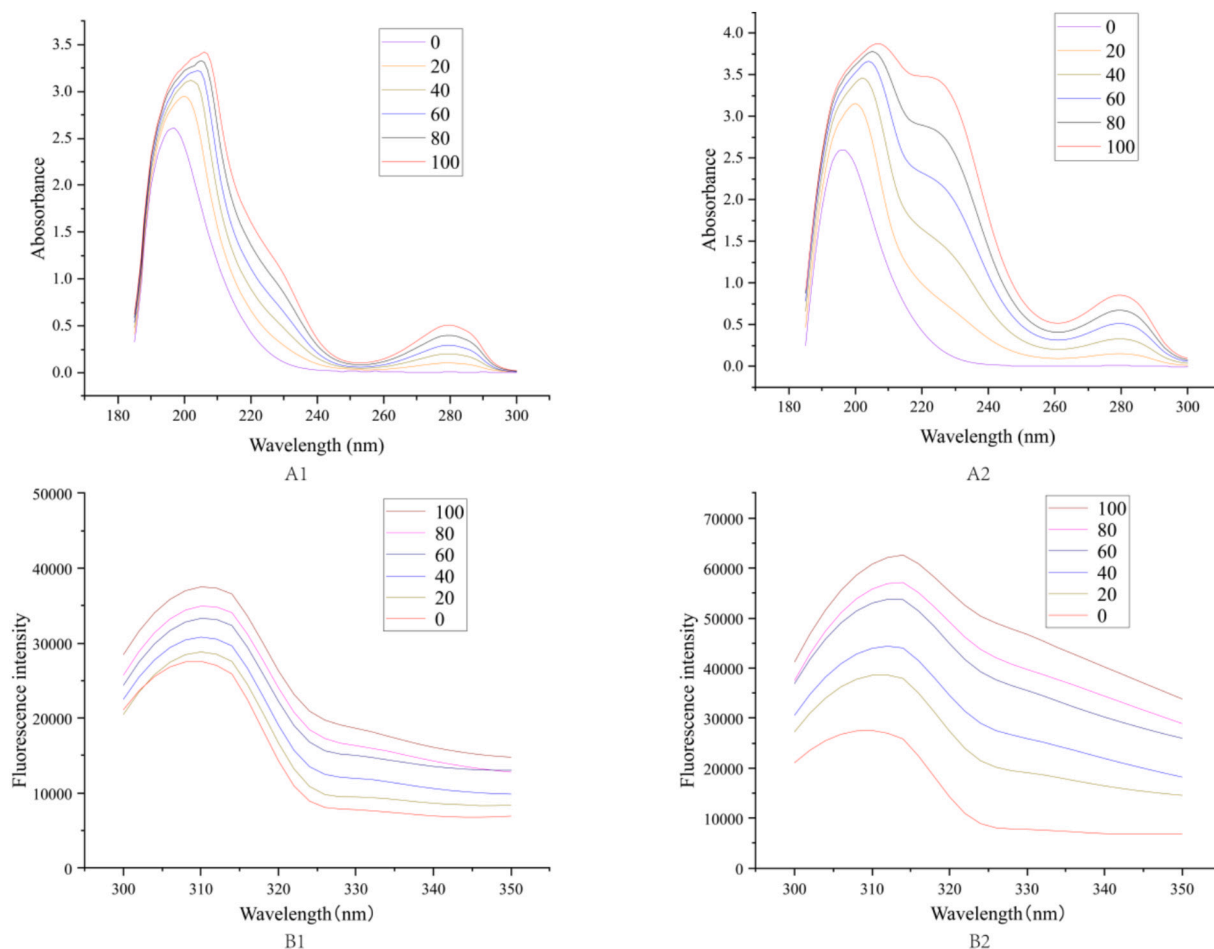


Fig. 5. UV-vis (A) and fluorescence (B) spectra of collagen, collagen-6-GOL, and collagen-GGA complexes. (A1–B1) 6-GOL; (A2–B2) GGA.

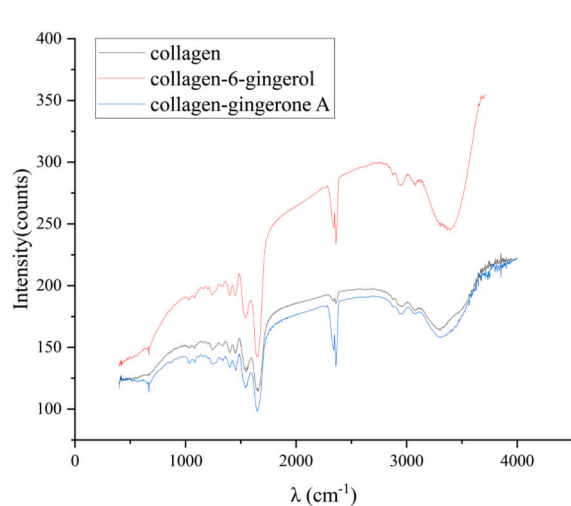


Fig. 6. FTIR spectra of collagen, collagen-GGA, and collagen-6-GOL complexes.

connections. The system maintained its stability because of its wide network of hydrogen bonds which linked its molecular structure (Alharbi, 2025). The SASA parameter showed how much of the protein molecular structure was accessible to solvent molecules while giving information about protein structural density (Shah et al., 2025). The SASA values of these three complexes showed a small increase when compared to apo collagen which indicated that the protein surface

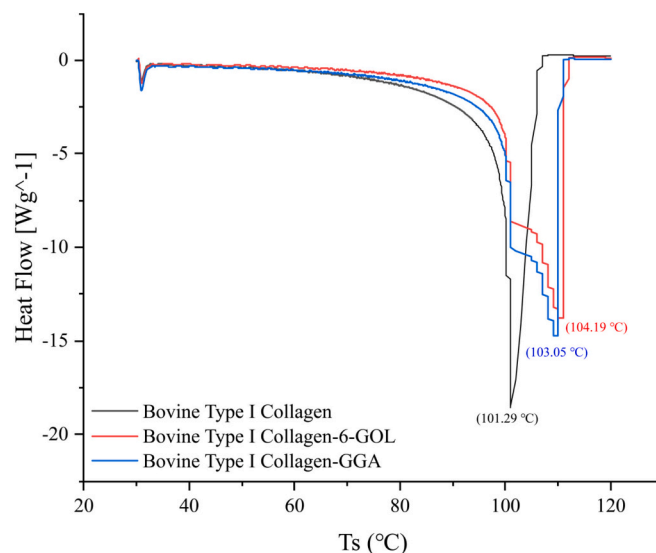


Fig. 7. DSC analysis of collagen, collagen-GGA, and collagen-6-GOL complexes.

became slightly more open (Du et al., 2016). The SASA values showed that GGA-collagen forms the most compact structure because it exhibited the smallest surface area among all the complexes. The SASA values for 6-GOL-collagen and 6-SHO-collagen show minimal growth which suggests that binding causes a small expansion of solvent-accessible areas (Yoosefian et al., 2025). The compounds in ginger strengthen

collagen structures through their ability to decrease protein movement while creating more stable protein arrangements (Zhuang et al., 2025).

The researchers validated their computational results through multispectral analysis by examining the collagen-6-GOL and collagen-GGA complexes. The addition of 6-GOL or GGA to collagen solutions produced a 205 nm peak which UV-vis absorption spectroscopy detected. This band is mainly attributed to electronic transitions of the peptide backbone (Tie et al., 2024). However, the enhanced concentration dependence of absorbance suggests that the microenvironment around aromatic amino acid residues undergoes changes (Jafari et al., 2020). The peptide backbone structure shows local changes because the spectrum shows a small red shift at 205 nm. (Maciazek-Jurczyk & Sulkowska, 2015). The opposite changes in absorbance at 280 nm observed for the small molecule and the complex (an increase for 6-GOL and a decrease for GGA) indicate that the spectral alterations are by no means a simple physical superposition of the intrinsic absorbance of the small molecules. Rather, it strongly confirms that an interaction has indeed occurred between collagen and the ligands (6-GOL and GGA), and that this binding has led to distinct changes in the chromophores' microenvironments (Gholizadeh et al., 2023). The FS study confirmed this process through its observation of collagen structure modifications which occur when the protein binds to ligands. The aromatic amino acids in collagen become excited by endogenous light which causes their aromatic chromophores to experience electronic transitions after they absorb photons (Dos Santos Rodrigues et al., 2023). The addition of 6-GOL or GGA resulted in a steady increase of fluorescence intensity at 309 nm which followed an opposite pattern to the typical fluorescence quenching that occurs when proteins experience structural degradation. This indicates alterations in the polarity or hydrogen bonding network of the microenvironment surrounding aromatic residues, consistent with the formation of hydrogen bonds and other interactions between the protein-ligand complex (Abeydeera et al., 2018). The fluorescence enhancement indicates that both ginger compounds bind to collagen structures while changing the surrounding environment through their ability to limit protein movement and create more hydrophobic conditions which enhance collagen stability. While we did not directly measure surface hydrophobicity using probes, it is hypothesized that the fluorescence enhancement indicates that both ginger compounds bind to collagen structures, limiting protein movement and creating more hydrophobic conditions.

The FTIR spectroscopy method allowed researchers to study how the secondary structure of the collagen-ligand complex changes. The spectrum exhibited Amide characteristic absorption bands sensitive to hydrogen bonding and conformational changes (Xiang et al., 2022). Compared to apo collagen, the collagen-6-GOL and collagen-GGA complexes exhibited intensity changes in Amide I and the Amide II (Wang et al., 2022). The spectral line displacement measured in these experiments provides direct evidence for computational prediction. Specifically, the structural displacement observed in Amide I (C=O stretching) and Amide II (N-H bending) is directly related to the intermolecular hydrogen bond network identified in molecular docking and molecular dynamics simulation, in which residues such as Arg179 and Tyr161 act as key functional anchors. The peptide shows modified C=O and N-H stretching vibrations through its intensity changes because the collagen-ligand complex forms new hydrogen bonds (He et al., 2022). The Amide I bands of 6-GOL and GGA complexes showed secondary structural changes, which is consistent with the maintenance of the characteristic amide spectrum and reflects the retention of the basic characteristics of the polypeptide main chain (Zheng et al., 2024). Additional spectral changes were detected in the 1000–1200 cm^{-1} region, potentially arising from C=O and C-C stretching vibrations. This further confirmed the interaction between the phenolic hydroxyl and carbonyl groups (Mir & Wilson, 2024; Munir et al., 2023). These results are consistent with previous reports on collagen-curcumin and collagen-polyphenol systems, where the red shift and intensity changes in the Amide band are attributed to the

rearrangement of hydrogen bond networks (Ghavimi et al., 2020; Shubina et al., 2024). The findings demonstrate that 6-GOL and GGA establish collagen stability through hydrogen bonding and hydrophobic interactions which confirms the potential of ginger compounds for meat processing functional stabilization.

To further verify the effects of 6-GOL and GGA on collagen stability, the thermal denaturation temperatures of pure collagen and the complexes formed between collagen and the two compounds were measured separately. The results showed that these two ginger-based compounds could increase the thermal denaturation temperatures of collagen to a certain extent. Research shows that thermal denaturation temperature is an important indicator to measure the structural stability of collagen, and the increase in temperature usually means the enhancement of structural stability (Jeong et al., 2023; Luo et al., 2025; Mao et al., 2024; Naryzhny et al., 2022; Xie et al., 2024). This confirms that the addition of both 6-GOL and GGA contributes to enhancing the structural stability of collagen.

Mechanistically, our multispectral and computational results indicate that ginger-derived compounds (6-GOL and GGA) bind collagen predominantly through hydrogen bonding and hydrophobic interactions, which is consistent with prior collagen-polyphenol studies showing that these two forces are the major contributors to polyphenol-mediated collagen stabilization (Cecchin et al., 2017). In line with reports that catechin treatment can increase the thermal stability of type I collagen (e.g., reduced shrinkage at high temperature), our DSC results similarly support that 6-GOL and GGA enhance collagen thermal stability. (Madhan et al., 2005). In addition, previous studies using FTIR/X-ray diffraction/X-ray photoelectron spectrum have confirmed that catenin-based cross-linking would trigger detectable structural changes in collagen and improve its overall physical properties (Mugaranja & Kulal, 2020). Accordingly, our multispectral analysis also provides evidence for the structural stabilization of collagen after interacting with 6-GOL and GGA. However, some limitations should be taken into account when interpreting these research results. Specifically, the present study relies on a predictive collagen model, which is not yet verified in the actual meat production system.

5. Conclusion

The research used computational and multispectral analysis techniques to study how ginger compounds modify bovine type I collagen. The computational studies revealed that Pro168, Tyr161 and Arg179 formed essential binding sites. The collagen-GGA/6-GOL complexes showed lower RMSD values and maintained their hydrogen bond structures throughout the MD simulation. Multispectral experiments corroborated these computational findings; the changes in the Amide A, B, I and II bands provide strong evidence for the strengthening of the intermolecular hydrogen bond network within the collagen complexes. Furthermore, DSC analysis demonstrated that the addition of both 6-GOL and GGA contributed to enhancing the structural stability of collagen. While these findings demonstrate that 6-GOL and GGA can serve as natural collagen structural stabilizers at the molecular level, direct extrapolations regarding improvements in meat texture, tenderness or water retention properties require caution. Future studies should focus on validating the stabilization effects and thermal mechanics of these ginger compounds within complex, collagen-rich meat matrices.

CRediT authorship contribution statement

Shao-Hua Wen: Writing – original draft, Methodology, Investigation, Formal analysis. **Hui-Ke Ma:** Writing – original draft, Investigation, Formal analysis. **Liang Shen:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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